Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/14/2015
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Using Brachypodium distachyon to Investigate Monocot Plant Adaptation to Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	53706-1580	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	09/12/2014	**End Date:**	09/11/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/11/2017 per NSSC information (Ed., 6/14/16)		
Key Personnel Changes/Previous PI:	No changes.		
COI Name (Institution):			
Grant/Contract No.:	NNX14AT23G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Most major cereal grain crops are monocots. Yet, most investigations of plant adaptation to the spaceflight environment have been carried out on the dicotyledonous model plant Arabidopsis thaliana. It remains unknown whether the conclusions from such studies can be extrapolated to monocotyledonous plants. To fill this knowledge gap, we propose to investigate the growth, development, and transcriptome profiles of Brachypodium distachyon seedlings under spaceflight conditions, and compare these adaptive responses to those recently documented for Arabidopsis seedlings. These investigations will lead to a better understanding of the molecular and developmental mechanisms that contribute to Brachypodium adaptation to spaceflight conditions. In the long term, these data may also lead to the development of better mitigation strategies aimed at improving monocot adaptability to spaceflight parameters.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Because most cultivated crops are monocotyledonous plants, there is tremendous interest in better understanding the molecular mechanisms that govern their growth behaviors and adaptation to stress. In addition to contributing to a better understanding of the molecular mechanisms that modulate plant adaptation to the microgravity environment, an important step toward engineering crops that are better suited to serve as components of the bioregenerated life-support systems during spaceflight, our work with Brachypodium distachyon should provide new insights into the molecular mechanisms that control root growth and development. Because root growth and its architecture have recently been recognized as key contributors to plant productivity and adaptation to environmental stress, our research is also likely to have long-term impact in agriculture, horticulture, and forestry. Furthermore, many of the behaviors under investigation involve auxin-related processes. This plant hormone is a key contributor to most aspects of plant growth, development, and response to the environment. Therefore, this project is likely to yield important new insights into some of the key mechanisms that control plant growth, development, and response to biotic and abiotic stresses.
Task Progress:	Most of the cultivated crops are monocotyledon plants. Yet, very little information is currently available on how monocots will respond to spaceflight conditions even though there is substantial interest in using them in bio-regenerative life support systems for long-term space exploration missions. Our research project is aimed at filling this knowledge gap by investigating the morphological and molecular adaptations to microgravity of Brachypodium distachyon seedlings, a model species for molecular genetic investigations of monocot development. We have been working toward defining the experimental parameters that will allow us to grow Brachypodium seedlings in the International Space Station (ISS), where their morphological and molecular adaptability to the microgravity environment can be evaluated. We are also mining the vast natural variation existing in this species to uncover some of the molecular mechanisms that govern monocot growth responses to gravity stimuli under 1g conditions, with the ultimate goal of testing their ability to contribute to microgravity adaptation. During this first year of our project, we have been able to better define the experimental parameters that will be needed to grow sterile Brachypodium distachyon seedlings in Petri dishes in the microgravity environment of ISS, within the VEGGIE chamber, for morphological and molecular comparisons with ground-based controls. A simple method for surface-sterilization of the seeds has been developed, which prevents microbial contamination of the cultures while maintaining full germination capacity. Medium composition and seed plating have been optimized to assure maintenance of medium integrity and prevent seeds from bouncing off the surface during spacecraft takeoff. We also demonstrated that application of blue-light during cold storage inhibits the germination of plated seeds for a period long enough to allow flight preparations, plates loading on the spacecraft, vehicle takeoff, travel to and docking with ISS, and material transfer to the growth chamber on ISS. Furthermore, a red-light treatment was developed to promote synchronized seed germination after removal from cold storage and transfer to the growth chamber. We also established the maximal period of seedling growth allowed considering the space constraints imposed by the dishes, and are in the process of optimizing protocols for seedling fixation at the end of the experiment, and RNA extraction for subsequent transcription profiling. These methods have been optimized for Bd21-3, an easily transformable and widely used Brachypodium accession. However, in the next few weeks, we will also test them on five other accessions, which we plan to incorporate in the microgravity-adaptation experiment because they display distinct root sensitivities to gravity and/or show distinct growth behaviors in response to gravistimulation on Earth. For all six accessions, plants have been propagated, generating more than enough seeds to carry out both microgravity and ground-based control experiments. We expect being able to complete a Science Verification Test at Kennedy Space Center (KSC) in December of 2015, as recently discussed with the KSC Support Team. In addition to this Definition phase of our project, we have developed more sophisticated approaches to analyze the gravisensitivities of our Brachypodium accessions, and have discovered a novel root-growth behavior associated with long-term clinorotation, which we hope to correlate with gravisensitivity. We have also collaborated with Dr. Nathan Miller in the Spalding laboratory (University of Wisconsin-Madison, Botany Department) to develop a high-definition image acquisition and analysis platform that allows extraction of defining spatio-temporal characteristics of Brachypodium root graviresponses. This analytical platform will be adopted to analyze all available Brachypodium accessions, and mathematical models will be fit to the data as a way to extract informative parameters defining each accession's gravi- and proprio-sensitivity. These parameters will then be introduced into Genome-Wide Association Study (GWAS) programs to identify novel loci that contribute to gravisensing and proprioception in this system.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Abstracts for Journals and Proceedings	Su S-H, Barker R, Masson PH. "Using Brachypodium distachyon to Investigate Monocot Responses to Microgravity and Gravistimulation." Presented at Midwest Plant Cell Dynamics Conference, Madison, WI, June 16-19, 2015. Midwest Plant Cell Dynamics Conference, Madison, WI, June 16-19, 2015. , Jun-2015
Abstracts for Journals and Proceedings	Su S-H, Barker R, Woods D, Masson PH. "Using Brachypodium distachyon to investigate monocot responses to gravistimulation and adaptation to spaceflight." Presented at Minisymposium 28: Space Biology. Plant Biology 2015, Minneapolis, Minnesota, July 26 - 30, 2015. Plant Biology 2015, Minneapolis, Minnesota, July 26 - 30, 2015. , Jul-2015
Articles in Peer-reviewed Journals	Su S-H, Gray WM, Masson PH. "Auxin: Shape matters." Nature Plants 2015 Jul;1(7):15097. Published online 07 July 2015. http://dx.doi.org/10.1038/NPLANTS.2015.97 , Jul-2015